LESSON PLAN

EMI/EMC (14)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	46	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Practical demonstration on Globe (Earth's Model) and how to do compass swing.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Practical demonstration on Globe (Earth's Model) and how to do compass swing.
5. Specific Objectives	14.1.1.1 Trainee will be able to explain and define a position on the earth.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Globe positioning and Compass Swing procedure with instructor supervision.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the shape of the Earth?
(b) Define magnetic north.
(c) Why is compass swing necessary?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Earth's Model Globe Representation • Latitude and Longitude Definition • Positioning on the Globe • Magnetic vs True North • Coordinate Systems	L & Ds	PPT, IB	15 min
(b)	• Compass Swing Fundamentals • Sources of Deviation • Swing Stand Setup • Cardinal Points Alignment • Deviation Coefficient Calculation	L & Ds	PPT, PDF	20 min
(c)	• Recording Deviation Cards • Interpolation Methods • Verification Procedures • Safety Precautions • Documentation Standards	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between magnetic heading and compass heading.
(b) List three sources of magnetic deviation.
(c) Describe the first step in setting up a compass swing.

12. Home Assignment and Follow Up	Prepare a brief note on safety precautions to be observed while working on compass and inertial navigation systems, referencing LP 09.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Safety precautions to be observed while working on compass, inertial navigation systems and safety precaution while working with nonmagnetic tools on compass.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min